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Applied cryptography and entropy source assessment

GOOD PRACTICE GUIDE · VERSION 4.1

libsp80090b: entropy assessment in practice

Good practice of SP 800-90B through the 18 lab cases, the restart test and conditioning

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ABSTRACT

This master-level guide teaches the min-entropy assessment of a noise source under **NIST SP 800-90B** [1], with **libsp80090b**, through the **18 cases** built by `lab/gen_data.sh`: healthy sources, design flaws, environmental effects, faults and outputs of deterministic generators. Each physical source is simulated from a model whose entropy is known, so every measurement can be checked against the truth, which no real source allows. Labs 3 and 4 complete the 18 cases with the restart test and conditioning.

The results come down to seven rules. Statistical tests do not see determinism: AES with an all-zero key scores 7.87 bits out of 8. An IID source can be poor: clipping in an ADC destroys 66 % of the entropy without creating any dependency. An estimate can exceed the truth: 3.43 bits measured for 3.33 real bits under a mains tone, hence the model's claim as a ceiling. An average hides the worst case: 3.38 bits credited for a source that ends at 2.38. Finally, only the restart test sees a source that always restarts from the same values. The guide ends with an eight-step qualification protocol, a complete worked example and the explanation of the computation times: test 1 can take an hour per file, because of 10,000 bzip2 compressions that only a failure forces to the end.

Keywords: libsp80090b; min-entropy; good practice; IID track; non-IID track; restart; conditioning; health tests; gen_data.sh; labs

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1 Purpose and how to use this guide

This guide is for assessors, cryptographers and engineers who qualify an entropy source. It assumes the SP 800-90B method (NT-01) and the use of the API and labs (NT-02). It does not compare the library with the NIST tool: that conformity is checked separately by `make check` (NT-02). It answers another question: **what should be concluded, and what should be done, in front of each result?**

Table 1. The four labs and the question each one asks.

Lab	Test (SP 800-90B)	Program	Question asked
1	IID track (§5)	`lab/lab1.sh`	Are the samples independent and identically distributed? If so, $H_{original}$ comes from the most common value estimator (MCV) alone.
2	Non-IID track (§6)	`lab/lab2.sh`	How much entropy is left against the ten estimators, predictors included? $H_I = \min(H_{original}, n \times H_{bitstring})$.
3	Restart (§3.1.4)	`lab/lab3.sh`	Does the source restart from an unpredictable state at every power-up?
4	Conditioning (§3.1.5)	`lab/lab4.sh`	How much entropy leaves a conditioning function, and when is the output full entropy?

The data come from `lab/gen_data.sh all --seed 90`: 10^6 samples per file, 1,000 × 1,000 for the restart test. The results are in `lab/results/labN_*/<case>.res`, with one summary line per case in `labN.log`.

> **Reading the numbers.** Another seed changes the estimates around the third decimal, and may flip an IID verdict close to the threshold (cases 3 and 17). The conclusions of this guide do not depend on these decimals.

2 Seven golden rules

Each rule is demonstrated by at least one of the cases of section 4.

1. **Assess the raw noise, never a conditioned output.** A DRBG or a hash passes every test, even with no entropy at all (cases 14 to 18). Capture upstream of any deterministic function.
2. **The physical model first, the measurement second.** The tests are a black-box check. An estimate can exceed the real entropy (case 8): the model's claim H_{claim} is the ceiling.
3. **The IID track must be earned.** It needs a physical argument **and** all three tests passed. A single rejected test forces the non-IID track (case 3), and a capture is not repeated until it passes (case 17).
4. **IID does not mean a good source.** A clipped or biased source is IID and poor (cases 4 and 7). Look at the histogram before the verdicts.
5. **One campaign per condition, keep the worst case.** A file that mixes regimes gives an average, not the worst case (cases 9 to 11). Capture each temperature, voltage and ageing state separately.
6. **Test the start-up.** A source that is healthy in steady state may restart from the same values at every power-up; only the restart test sees it (lab 3).
7. **Apply the minimum rule, then size.** $H = \min(H_{claim}, H_I, H_r, H_c)$; the seed, the health test cutoffs and the conditioning margin follow from H (section 6).

3 Map of the 18 cases

Table 2 summarises the 18 cases of labs 1 and 2, in the order of `gen_data.sh --list`. The "Expected H" column is the min-entropy per sample given by the model; lab 1 gives the IID verdict and, in brackets, $H_{original}$, which is only usable if the verdict is IID.

Table 2. The 18 cases: model, expected entropy and results of labs 1 and 2.

No.	Dataset	Bits	Model (`gen_data.sh`)	Expected H	Lab 1: verdict ($H_{original}$)	Lab 2: H_I	Lesson
1	thermal	8	Gaussian noise σ = 20 on an 8-bit ADC	5.65	IID (5.618)	4.093	Reference; cost of the non-IID track
2	thermal-lsb	1	Least significant bit of the same ADC	1.00	IID (0.992)	0.922	Keep the bits the noise really moves
3	ringosc-slow	1	Oscillator, 0.5 period of jitter between two reads	≈ 1	not IID (0.996)	0.861	One rejected test is enough
4	saturated	8	Gain too high (σ = 200), clipped to 0 and 255	1.93	IID (1.921)	1.727	IID but poor
5	lowpass	8	First-order filter, α = 0.95	3.97 [a]	not IID (5.626)	3.248	Oversampling
6	ringosc-fast	1	Oscillator read too fast (0.08 period per read)	low	not IID (0.996)	0.110	MCV is blind to runs
7	biased-1bit	1	Comparator offset: P(1) = 1/4	0.415	IID (0.414)	0.250	A counted bias is not a flaw
8	interference	8	40-step sine, period 20, over noise σ = 4	3.33 [b]	not IID (5.726)	3.430	The estimate exceeds the truth
9	aging	8	σ falls from 20 to 2 along the file	5.65 → 2.38	not IID (4.275)	3.381	An average is not the worst case
10	warmup	8	σ = 0.3 over the first 20 %, then σ = 20	1.00 then 5.65	not IID (3.094)	1.019	Start-up ruins the whole file
11	drift	8	Uniform 4 bits, then uniform 8 bits	4 then 8	not IID (4.884)	2.440	Assess each regime
12	stuck	8	One symbol only (`/dev/zero`)	0	refused (−3)	refused (−3)	Handle the error codes
13	replay	8	One 4 KiB block repeated	≈ 0 [c]	no verdict (−4)	1.6×10^{-6}	Health tests do not see a replay
14	urandom	8	Kernel DRBG output	out of scope [d]	IID (7.882)	7.323	Ceiling of the estimators
15	urandom-4bit	4	Same output folded to 4 bits	out of scope	IID (3.972)	3.539	Ceiling of the estimators
16	urandom-1bit	1	Same output folded to 1 bit	out of scope	IID (0.994)	0.857	Ceiling for a binary source
17	openssl	8	OpenSSL DRBG output	out of scope	not IID (7.896) [e]	6.638	Spread of the estimates
18	drbg-zero-seed	8	AES-256-CTR, all-zero key and IV	0	IID (7.866)	7.145	No test sees determinism

[a] Entropy conditioned on the past: the filter innovation has a standard deviation of $20 \times \sqrt{(1 - 0.95^2)} = 6.24$. [b] The sine is deterministic: an attacker knows its phase. [c] The whole file holds at most 4,096 × 8 = 32,768 bits of entropy, those of the first block. [d] A DRBG output only has the entropy of its seed; SP 800-90B does not apply. [e] False rejection: a fresh `openssl rand` file passes all three tests.

4 Analysis of the cases

4.1 Reading a lab 2 output

On an n-bit source, `test_non_iid -vv` applies the estimators to the symbols ("Literal") and to the bit string ("Bitstring"). $H_{original}$ is the minimum of the former, $H_{bitstring}$ that of the latter, and $H_I = \min(H_{original}, n \times H_{bitstring})$. For case 1: the weakest literal estimator is the t-tuple (5.492), the weakest on bits is compression (0.5116), so $H_I = \min(5.492; 8 \times 0.5116) =$ **4.093**. Identifying the estimator that sets H_I is the first step of the diagnosis: it tells which structure was found.

Table 3. Weakest estimators, and what they reveal.

Weakest estimator	Cases	What it found
Compression (bits)	thermal, aging, drift, interference, urandom, urandom-4bit, drbg-zero-seed	Nothing specific: a cautious bound, which costs a healthy source about 1.4 bits out of 8
MultiMCW (symbols)	openssl (6.638)	A local regularity, due to chance
Lag predictor	lowpass (3.976 on the symbols)	The correlation with the previous sample
MultiMMC (bits)	lowpass (0.406 × 8 = 3.248)	The transitions of the bit string
Longest repeated substring (LRS)	replay, warmup	Long repeated sequences
Compression (symbols)	biased-1bit, ringosc-fast, urandom-1bit	The runs and the bias of a binary source
Collision	ringosc-slow, saturated (bits)	Close repetitions
t-tuple	thermal-lsb	Frequent patterns

4.2 Healthy sources: the reference (cases 1 to 3)

Case 1, thermal. The model gives 5.65 bits; lab 1 concludes IID with 5.618 bits, the 99 % bound of the MCV costing only 0.03 bit. The non-IID track credits 4.093 bits: **1.5 bits less, i.e. 27 %**, the price of having no IID argument. This is the value of a sound physical model: if the model justifies IID, 5.618 bits are credited; otherwise only 4.093.

Case 2, thermal-lsb. The least significant bit of a noise of σ = 20 steps is perfectly balanced: IID at 0.992, non-IID at 0.922, on a par with a perfect binary source (case 16: 0.857). Per captured bit, this is much denser than the full byte (4.093 / 8 = 0.51 bit per bit). Per sample, the byte remains richer: keeping only the bits the noise moves is a throughput choice, not an obligation.

Case 3, ringosc-slow. The phase accumulates 0.5 period of jitter between two reads: the source is nearly IID. It passes the chi-square and longest repeated substring tests, but the statistic counting runs around the median rejects the hypothesis (C_0 = 9,998 out of 10,000). The rejection is sound: a bit equals the previous one with probability 0.5021 instead of 0.5, i.e. 4.2 standard deviations over 10^6 bits, a likely trace of the regular drift of 0.08 period per read. **One rejected test is enough**; the non-IID track credits 0.861 bit.

4.3 Design flaws (cases 4 to 8)

Case 4, saturated. A gain that is too high sends 26 % of the samples to 0 and 26 % to 255. The source remains IID (1.921 bits, 1.928 in theory), but has lost **66 %** of the entropy of case 1. Clipping destroys entropy without creating dependency: no independence test can report it. Only the histogram and the comparison with the model show it. Fix: reduce the analogue gain so that the noise fits within the ADC range.

Case 5, lowpass. Sampling faster than the noise bandwidth correlates two neighbouring samples (α = 0.95). All three IID tests fail. The Lag predictor finds **exactly** the conditional entropy of the model (3.976 measured for 3.97), and H_I = 3.248 stays below. The MCV alone would have given 5.626 bits: crediting $H_{original}$ without an IID verdict would have overestimated the source by 2.4 bits. Fix: lower the sampling rate, or keep one sample out of k, until the autocorrelation is negligible.

Case 6, ringosc-fast. The oscillator moves 0.08 period per read: the level only changes about every six samples, and a bit equals the previous one 84 times out of 100. The bits remain balanced: the MCV gives 0.996, but compression finds 0.110. **An unjustified IID track would overestimate by a factor of 9.** Fix: divide the sampling clock so that the jitter accumulated between two reads covers at least one period.

Case 7, biased-1bit. P(1) = 1/4: theory gives $-\log_2(3/4) = 0.415$ bit, lab 1 concludes IID at 0.414. A bias is not a flaw if it is counted: the seed is simply longer. The non-IID track is very cautious here (0.250, compression estimator): with 384 bits to collect, the IID track needs 929 samples and the non-IID track 1,538. A debiaser (von Neumann) placed before the capture point is part of the noise source and must be modelled; placed after it, it is not a vetted conditioning function.

Case 8, interference. A 40-step mains tone dominates a noise of σ = 4. For an attacker who knows the phase, the entropy is that of the noise alone: **3.33 bits**. Yet $H_I = 3.430$: **the estimate exceeds the truth by 0.1 bit**, because none of the ten estimators models a sine. This is why the minimum rule exists: H_{claim}, from the model, caps the estimate. Fix: decouple the supply and shield; the periodic component could moreover be injected by an attacker.

4.4 Environment and non-stationarity (cases 9 to 11)

Case 9, aging. The noise fades from σ = 20 to σ = 2 along the file. $H_I = 3.381$ is an **average**: at the end of the file, the source is only worth 2.38 bits. An APT set for 3.381 bits (W = 512, C = 85) only fires on average below about 2.6 bits: in between, the source delivers less than it is credited with, without any alarm. Good practice: qualify on aged parts and at the extremes of temperature and voltage, one file per condition, and keep the minimum.

Case 10, warmup. The first 20 % of the samples take only four values, with 1.00 bit. The longest repeated substring finds them and sets $H_I = 1.019$ **for the whole file**: the start-up phase has cost the steady state 3 bits. Good practice: determine the settling time, discard those samples at start-up, capture the campaign in steady state, and check the start-up with the restart test.

Case 11, drift. The distribution changes halfway: 4 uniform bits, then 8. The worst regime is worth 4 bits; $H_I = 2.440$, because the bit estimators see the four most significant bits at zero in half of the file. The result is cautious but describes no real regime. Good practice: find the breaks (mean and histogram per slice) and assess each regime separately.

4.5 Hard faults (cases 12 and 13)

Case 12, stuck. One symbol only: both tracks refuse the file with the code `SP80090B_EALPHABET` (−3). An error code is not a zero result to be interpreted later: the caller must stop the qualification. In operation, the RCT detects this fault within a few samples.

Case 13, replay. A 4 KiB block is read again and again, like a DMA buffer that is no longer refreshed. The longest repeated substring sets $H_I = 1.6 \times 10^{-6}$ bit: the fault is detected. The IID track fails the chi-square test, then the longest repeated substring test cannot compute its probability, too small for a `long double`: code `SP80090B_EINTERNAL` (−4), no verdict (limitation L1). A major lesson: **health tests do not see a replay**. The replayed block looks random: no consecutive repetition for the RCT, no abnormal proportion for the APT. The countermeasure lies in the design: frame counter, timestamp or buffer freshness check.

4.6 Conditioned outputs: the trap (cases 14 to 18)

Case 18, drbg-zero-seed. AES-256 in counter mode with an all-zero key and IV: the sequence is entirely predictable, its entropy is zero. Yet it passes all three IID tests with 7.866 bits, and the non-IID track grants it 7.145 bits.

> **No statistical test sees determinism.** The estimators measure predictability through simple models; a cryptographic algorithm fools them all. The assessor must require proof that the samples come from the physical component, upstream of any deterministic function: schematic, capture point and acquisition code.

Cases 14 to 16, urandom. These DRBG outputs must not be assessed as sources; they serve as a **reference of perfect randomness** and show the ceiling of the estimators over 10^6 samples: 7.88 bits on the IID track and

about 7.3 on the non-IID track for 8 bits, 3.54 for 4 bits, 0.86 for 1 bit. No real source can be credited beyond that.

Case 17, openssl. Two lessons. First, the IID track rejects this file on the periodicity statistic at lag 8 alone (C_2 = 9,997): this is a **false rejection**, which happens to about 2 to 3 % of IID sources, and a fresh file passes. The standard still imposes the non-IID track on the rejected file: repeating a capture until it passes would amount to choosing one's result. Second, the non-IID track gives 6.638 bits, against 7.323 for urandom: the MultiMCW predictor found a local regularity by chance. **Two perfect sources differ here by 0.7 bit**: an estimate is a random variable, and a margin on H is prudent.

5 Restart and conditioning

5.1 Lab 3: the restart test

The restart test (§3.1.4) arranges 1,000 power-ups of 1,000 samples in a matrix. It first checks that no value dominates a row or a column ($X_{max} \le X_{cutoff}$), then estimates the entropy of the rows (H_r) and of the columns (H_c); $\min(H_r, H_c) \ge H_I / 2$ is required.

Table 4. Lab 3: restart test (1,000 × 1,000 samples).

Dataset	H_I (lab 2)	X_{max} / X_{cutoff}	H_r / H_c	Validated H	Result
restart-thermal	4.093	39 / 99: passed	5.306 / 5.306: passed	**4.093**	**Qualified**
restart-urandom	7.323	15 / 24: passed	7.719 / 7.354: passed	**7.323**	**Qualified**
restart-warmup	4.093	1,000 / 99: **failed**	—	0	**Rejected**
restart-frozen	7.323	1,000 / 23: **failed**	—	0	**Rejected**

- **The fault invisible in continuous operation.** restart-warmup has exactly the distribution of case 1 in steady state, but its first 20 samples are the same at every start-up: a whole column is constant (X_{max} = 1,000). No continuous capture reveals it.
- **X_{cutoff} depends on H_I.** The higher the claimed entropy, the lower the cutoff (99 for 4.09 bits, 23 or 24 for 7.32 bits); it is simulated, hence the difference of one for the same H_I.
- **H_I remains the bound.** H_r and H_c exceed H_I on the healthy sources: the validated value remains H_I. The test only confirms, or cancels, the entropy of lab 2.
- **Good practice:** real power-ups (supply cut), not software restarts; the source must have reached the state in which it will deliver its first seed.

5.2 Lab 4: conditioning and full entropy

Table 5. Lab 4: conditioning to a 256-bit output ($n_w = 256$).

Case	n_{in}	h_{in}	h'	h_{out} (bits)	Full entropy
vetted-320-margin	320	320	—	$256 - 6 \times 10^{-9}$	**Yes**: 64-bit margin
vetted-512-6.4	512	409.6	—	256	**Yes**
vetted-seed-384	3,072	384	—	256	**Yes**
vetted-thermal	632	323.3	—	256	**Yes**: 79 samples of case 1
vetted-256-full	256	256	—	251.69	**No**: no margin
vetted-128-short	256	128	—	128	**No**: input too short
nonvetted-0.99	512	409.6	0.99	253.44	**No**: ceiling $h' \times n_{out}$
nonvetted-urandom	512	409.6	0.915	234.34	**No**
nonvetted, 4 others	512	409.6	0.857 to 0.908	219.38 to 232.38	**No**

- **The 64-bit margin.** With a vetted function (SHA-256), 256 bits of input entropy give 251.69 bits of output: the collisions of the function cost 4.3 bits. Full entropy requires $h_{in} \geq n_{out} + 64 = 320$ bits [3] [4].
- **The ceiling of a non-vetted function.** h_{out} never exceeds $h' \times n_{out}$, h' being the entropy per bit measured on the output of the function. Even perfect randomness gives h' between 0.86 and 0.92 over 10^6 samples (cases 14 to 18): a non-vetted function never gives full entropy.
- **From lab 2 to lab 4.** vetted-thermal turns the measurement into a number of samples: $\lceil 320 / 4.093 \rceil = 79$ samples, i.e. 323.3 bits, for a full-entropy SHA-256 output.

6 Qualification protocol

Figure 1 arranges the golden rules into a protocol; the functions named are those of the API (prefix `sp80090b_`).

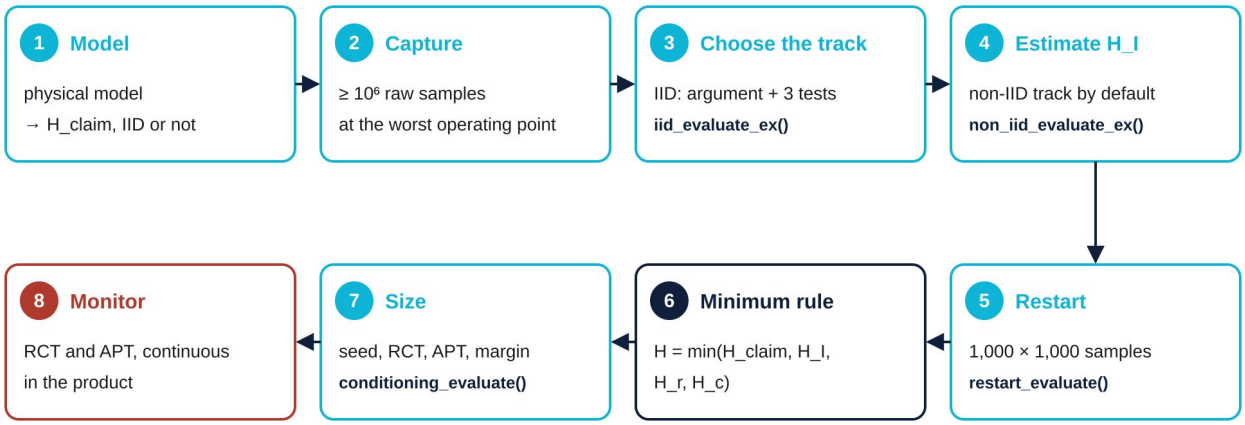


Figure 1. Qualification protocol of a noise source with libsp80090b.

1. **Model.** Describe the physical phenomenon, derive H_{claim} from it and state whether IID can be justified.
2. **Capture.** At least 10^6 consecutive raw samples, at the documented capture point, for each extreme condition (temperature, voltage, ageing); discard the start-up phase.
3. **Choose the track.** IID track only with the model's argument and all three tests passed on the first file; otherwise the non-IID track.
4. **Estimate H_I.** Check the return code, note the weakest estimator, compare with H_{claim}; a large gap in either direction calls the model into question.

5. **Restart.** 1,000 real power-ups × 1,000 samples.
6. **Apply the minimum rule.** H = min(H_{claim}, H_I, H_r, H_c), on the worst condition.
7. **Size.** Seed, RCT and APT cutoffs, conditioning input.
8. **Monitor.** RCT and APT on every raw sample, in the product; at the first failure, discard the current seed.

Table 6. Complete example: sizing for the thermal noise (case 1), H = 4.093 bits.

Quantity	Computation	Value
H_I (lab 2)	min(5.492; 8 × 0.5116)	4.093 bits
Restart (lab 3)	$H_r = H_c = 5.306 \geq H_I / 2$	validated
Retained H	min(H_I, H_r, H_c), no IID claim	**4.093 bits**
RCT, cutoff C	$1 + \lceil 20 / H \rceil$	**6**
APT, window and cutoff	W = 512 (non-binary source), $\alpha = 2^{-20}$	**C = 59**
Seed of a 256-bit DRBG	$\lceil 384 / H \rceil$	**94 samples**
Full-entropy SHA-256 output	$\lceil (256 + 64) / H \rceil$	**79 samples**

The APT window is 1,024 samples for a binary source and 512 otherwise (§4.4.2). With an IID claim justified by the model, H would be 5.618 bits, subject to a restart test run with that value.

7 Common mistakes

Table 7. Common mistakes, the cases that demonstrate them and good practice.

Mistake	Shown by	Good practice
Assessing a DRBG, hash or cipher output	Cases 14 to 18	Capture the raw noise, upstream of any deterministic function
Crediting $H_{original}$ without an IID verdict	Cases 5, 6, 8	$H_{original}$ is only valid if all three tests pass; otherwise H_I
Repeating the capture until IID passes	Cases 3, 17	The first file decides; a rejection imposes the non-IID track
Judging a source's quality by its IID verdict	Cases 4, 7	Compare H with the model, look at the histogram
Trusting the estimate over the model	Case 8	Cap with H_{claim}
Mixing several conditions in one file	Cases 9, 11	One file per condition, keep the minimum
Including the start-up phase in the campaign	Case 10, restart-warmup	Discard at start-up; restart test
Treating an error code as a result	Cases 12, 13	Stop and diagnose
Relying on RCT and APT to detect everything	Cases 9, 13	Health tests detect hard faults, not a replay or a slow degradation
Conditioning without a margin	vetted-256-full	$h_{in} \geq n_{out}$ + 64 bits
Reading three decimals as a certainty	Cases 14, 17	Keep a margin; several captures if possible

8 Computation time

Figure 2 gives the run times measured per file of 10^6 samples on an Intel Core i7-8650U (4 cores, 8 OpenMP threads). Three labs take seconds; test 1 can take an hour per file.

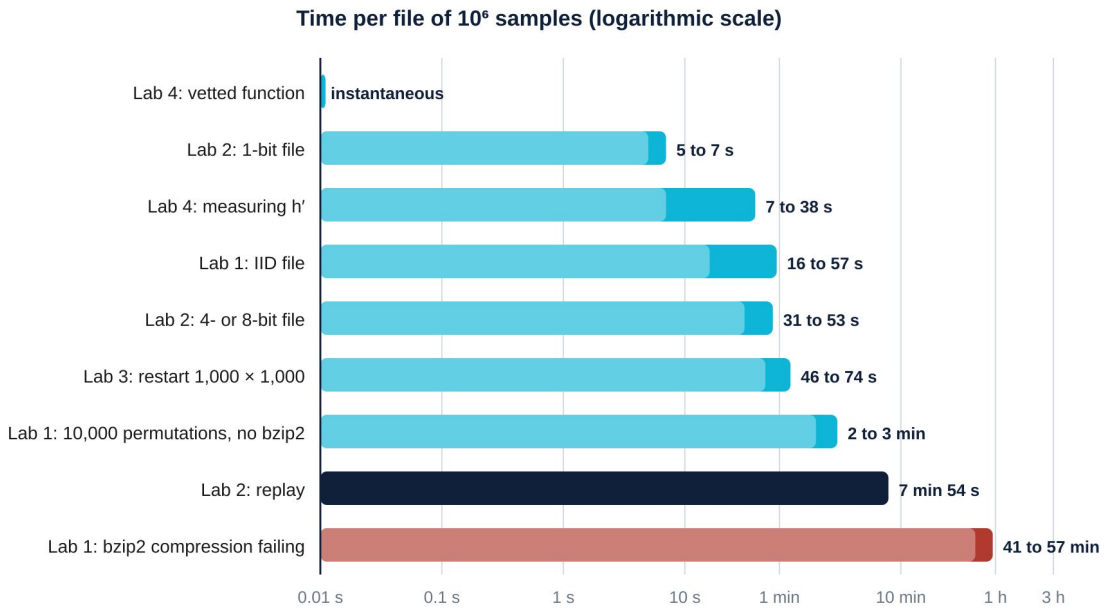


Figure 2. Run times measured per file of 10^6 samples (Intel Core i7-8650U, 8 threads).

8.1 Why test 1 is slow

Test 1 (§5.1) compares 19 statistics of the original data with the same statistics computed on **10,000 random permutations** of the data. If the samples are IID, the order does not matter and the original value ranks among the others; if it is among the 6 largest or the 6 smallest, the IID assumption is rejected. Three factors drive the run time:

1. **Volume.** Each permutation shuffles the whole file (10^6 samples) and recomputes the statistics still in play: up to 10^{10} samples processed per statistic.
2. **bzip2 compression.** The 19th statistic (§5.1.11) is the length of the bzip2 compression of the values written in decimal, about 4 MB of text for an 8-bit file. One compression takes 0.3 to 0.45 s on one core of the test machine: 10,000 compressions alone amount to nearly an hour of computation, the order of magnitude of the 41 to 57 min measured. The other 18 statistics are single-pass counts, much cheaper.
3. **The asymmetry between pass and fail.** The verdict of a statistic is settled as soon as the counters $C_0 + C_1$ and $C_1 + C_2$ both exceed 5: the library then stops computing it. On an IID source, this happens after a few tens to a few hundred permutations. A **failure**, however, is only decided after all 10,000: all permutations must be seen to assert that the original value stays among the 6 extremes.

Table 8 shows this on the lab 1 files: the run time depends less on the verdict than on the compression statistic.

Table 8. Lab 1: permutations computed and run time, per file.

Dataset	Verdict	Permutations (max.)	Permutations of the compression	Run time
thermal	IID	217	82	36 s
thermal-lsb	IID	3,750	16	57 s
openssl	not IID (periodicity(8))	10,000	76	under 3 min
ringosc-slow	not IID (numRunsMedian)	10,000	29	under 3 min
interference	not IID (18 statistics)	10,000	10,000	41 min
lowpass	not IID (18 statistics)	10,000	10,000	51 min
warmup	not IID (10 statistics)	10,000	10,000	53 min
ringosc-fast	not IID (18 statistics)	10,000	10,000	57 min

openssl and ringosc-slow run all 10,000 permutations, but their compression stops before the hundredth: under 3 min. As soon as the compression fails, the file costs nearly an hour.

8.2 Why the other tests are fast

- **Test 2 (5 to 53 s).** The ten estimators read the file once, or sort its suffixes once (longest repeated substring, t-tuple). An 8-bit file is assessed twice, on the symbols and then on the 8×10^6 bits, hence 31 to 53 s against 5 to 7 s for 1 bit. Exception: replay (8 min), because the repetition estimators go through every length up to the longest repetition, here almost the whole file.
- **Test 3 (46 to 74 s).** Two non-IID assessments (rows and columns of the matrix) and the simulation of X_{cutoff}.
- **Test 4.** A formula, instantaneous, for a vetted function; 7 to 38 s to measure h′ on the output of a non-vetted function (one test 2 assessment).

8.3 Practical consequences

- **Start with test 2.** In a minute, it gives H_I, valid in every case.
- **Run test 1 only if the model claims IID.** Without such a claim, the non-IID track applies and test 1 is useless; this is the default case of SP 800-90B.
- **Read the run time as a clue.** A test 1 that lasts more than a few minutes has at least one unsettled statistic, most often the compression: the verdict will almost always be "not IID". The end must still be awaited to conclude.
- **Budget the campaign.** The 18 files need 5 to 6 h in lab 1 (the six files whose compression fails account for most of it), against a quarter of an hour in lab 2: run lab 1 overnight, one lab at a time (limitation C2).
- **Do not shorten the file to save time.** The run time is proportional to the number of samples, but SP 800-90B requires at least 10^6 samples (§3.1.1); a shorter file distorts the confidence bounds.

9 Tool limitations to know

Table 9. Limitations of libsp80090b 2.0.0 that matter to the assessor.

Id	Limitation	What to do
L1	On a source that replays a long block, the longest repeated substring test of the IID track returns −4 instead of a verdict	Treat the source as not IID; the non-IID track gives the right result (case 13)
C2	Each call starts its own OpenMP team: simultaneous assessments oversubscribe the processor	Run one lab at a time, or set `OMP_NUM_THREADS`
M2	A few allocations of the MultiMMC and LZ78Y estimators and of the chi-square tests are not checked	Provide the memory needed; an abrupt stop means memory ran out

10 Conclusion

The 18 cases show what the SP 800-90B tests can do: detect dependencies (cases 3, 5, 6, 8), changes of distribution (cases 9 to 11), repetitions (case 13) and dead sources (case 12), and cautiously quantify the remaining entropy. They also show what they cannot do: see determinism (case 18), judge the quality of an IID source (case 4), guarantee a value no higher than the truth (case 8) or describe the worst case of a file that mixes regimes (case 9).

libsp80090b provides the measurements; security comes from the approach around them: a physical model, a capture at the right place and at the worst operating point, a restart test, the minimum rule and health tests set

on the retained value. This approach prepares an ESV submission [5]; it does not replace the NIST assessment.

11 References

[1] M. Sönmez Turan, E. Barker, J. Kelsey, K. McKay, M. Baish, M. Boyle. *Recommendation for the Entropy Sources Used for Random Bit Generation*. NIST SP 800-90B, January 2018.

[3] E. Barker, J. Kelsey, K. McKay, A. Roginsky, M. Sönmez Turan. *Recommendation for Random Bit Generator (RBG) Constructions*. NIST SP 800-90C, September 2025.

[4] National Institute of Standards and Technology. *Security Requirements for Cryptographic Modules*. FIPS PUB 140-3, 2019; Implementation Guidance, D.K.

[5] NIST CMVP. *Entropy Source Validation (ESV) Program*. ESV program documentation.

[7] J. Kelsey, K. A. McKay, M. Sönmez Turan. *Predictive Models for Min-Entropy Estimation*. CHES 2015, LNCS 9293.

Reference numbers are those of note NT-01.